

Safety Data Sheet



RCC Corrosion Control / SDS #: RCC-2100 / Revision Date: 05/23/2022

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Product Name: HARDENER NO. 2 Various Colors

Chemical Family: Organic / Ketone Peroxides

Product Use: Coating / Polymerization Catalyst

Manufacturer: RCC Corrosion Control
1450 Hoff Industrial Drive
O'Fallon, MO 63366
Phone: 636-697-4659

24-Hour Emergency Phone Number: North America: 800-424-9300 (CHEMTREC)
International: 703-527-3887 (CHEMTREC) Collect Calls Accepted

2. HAZARD IDENTIFICATION

GHS Classifications

Health Hazards

Acute Toxicity, Oral, Category 5
Acute Toxicity, Dermal, Category 4
Acute Toxicity, Inhalation, Category 3
Skin Corrosion, Category 1B
Serious Eye Damage, Category 1
Specific Target Organ Systemic Toxicity, Repeated Exposure, Category 2, Respiratory Tract [Inhalation, Ingestion, Skin absorption]
Aspiration Hazard, Category 2

Physical Hazards

Organic Peroxide, Type D
Corrosive to Metals, Category 1

Environmental Hazards

Acute Aquatic Toxicity, Category 2

GHS Labeling

Pictograms:

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Signal Word: Danger!

Hazard Statements

H242: Heating may cause a fire
H290: May be corrosive to metals
H302: Harmful if swallowed
H305: May be harmful if swallowed and enters airways
H312: Harmful in contact with skin
H314: Causes severe skin burns and eye damage
H331: Toxic if inhaled
H373: May cause damage to organs prolonged or repeated exposure
H401: Toxic to aquatic life

Precautionary Statements

Prevention:

P201: Keep away from heat/sparks/open flames/hot surfaces. No smoking.
P233: Keep container tightly closed.
P234: Keep only in original container.
P235: Keep cool.
P260: Do not breathe fume/mist/vapors/spray.
P264: Wash hands and exposed areas thoroughly after handling.
P271: Use only outdoors or in a well-ventilated area.
P280: Wear protective gloves/protective clothing/eye protection/face protection.

Response:

P301 + P330 + P331: IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
P302 + P352 + P310: IF ON SKIN: Wash with plenty of soap and water. Immediately call a POISON CENTER or doctor/physician.
P303 + P361 + P353: IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
P304 + P340 + P311: IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/physician.
P305 + P351 + P338 + P310: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/physician.
P312: Call a POISON CENTER or doctor/physician if you feel unwell.
P321: Specific treatment found in supplemental First Aid section of this SDS (Section 4).
P363: Wash contaminated clothing before reuse.
P390: Absorb spillage to prevent material damage.

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Storage:

P403 + P233: Store in a well-ventilated place. Keep container tightly closed.

P405: Store locked up.

P406: Store in corrosive resistant container with a resistant inner liner.

P411 + P235: Store at temperatures not exceeding 38°C (100°F). Keep cool.

P420: Store away from other materials.

Disposal:

P501: Dispose of contents/container in accordance with local, regional, and federal regulations

3. COMPOSITION/ INFORMATION ON INGREDIENTS

Chemical characterization

Organic / Ketone Peroxides

Component*	CAS #	Weight %
Dimethyl phthalate	131-11-3	35 - 60
2-Butanone Peroxide	1338-23-4	28 - 38
2-Butanone	78-93-3	1 - 3
Hydrogen Peroxide	7722-84-1	< 1

4. FIRST AID MEASURES:

Inhalation

Symptoms & Effects: Nose, throat and lung irritation, respiratory tract irritation, nausea, headache, dizziness, and drowsiness.

Measures: Remove victim to fresh air and keep at rest in a position comfortable for breathing. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Call a POISON CENTER or doctor/physician.

Skin Contact

Symptoms & Effects: Skin irritation, redness, burning sensation, drying, severe skin damage and corrosive damage to the skin.

Measures: Immediately wash skin with soap and plenty of water for at least 15 minutes. Remove contaminated clothing and shoes while washing. Immediately call a POISON CENTER or doctor/physician. Wash contaminated clothing before reuse.

Eye Contact

Symptoms & Effects: Eye irritation, strong eye pains, tearing, redness, burning sensations, swelling of the eyes, severe eye damage and blindness.

Measures: Immediately rinse eyes with water for at least 15 minutes. Remove contact lenses after the initial few minutes and if easy to do so and resume rinsing. Rinse beneath eyelids by holding eyelids apart with clean fingers while rinsing. Immediately call a POISON CENTER or doctor/physician.

Ingestion

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Symptoms & Effects: Stomach or intestinal irritation, nausea, vomiting, throat irritation, dizziness, drowsiness, headache, and weakness.

Measures: If swallowed, seek medical attention. Do not induce vomiting unless directed to do so by medical personnel. Victim should rinse mouth thoroughly if conscious and easy to do so. If the victim is drowsy or unconscious, do not give anything by mouth. Place individual on their left side and place their head down. Do not leave victim unattended.

5. FIRE FIGHTING MEASURES

Flash Point: 203°F (95°C)

Self-Accelerating Decomposition Temperature (SADT): 167°F (75°C) (45 gallon container)

Suitable Extinguishing Media: Water spray, Foam, Dry chemical

Unsuitable Extinguishing Media: None Identified

Hazardous Combustion Products: Hazardous organic compounds, Carbon monoxide, Carbon dioxide

Protective Equipment for Fire-Fighters: Wear full firefighting turn-out gear (full Bunker gear), and respiratory protection (SCBA).

Precautions for Fire-Fighters: Fight fire with large amounts of water from a safe distance. Closed containers of this material may explode when subjected to heat from surrounding fire. Cool closed containers exposed to fire with water spray. Contact with incompatible materials or exposure to temperatures above the SADT may result in self-accelerating decomposition reaction with release of flammable vapours which may auto-ignite. After a fire, wait until the material has cooled to room temperature before initiating clean-up activities.

6. ACCIDENTAL RELEASE MEASURES:

Protective Equipment: Recommended to wear chemical splash goggles & resistant gloves and discard of gloves that show tears, pinholes, or signs of wear. Wear proper garments to prevent skin exposure, such as long-sleeves and pants.

Personal Precautions: Persons not wearing proper PPE should be excluded from the area of contamination until clean-up has been completed. Ensure adequate ventilation. Pay close attention to the spreading of gases, especially at ground level.

Environmental Precautions: Do not allow discharge into drains, surface waters, or sanitary sewer system. Prevent spreading over a wide area by containment or oil barriers. Local authorities should be advised if significant spillages cannot be contained or if material discharges into drains or ground water.

Methods & Materials for Clean-Up: Eliminate all ignition sources. Avoid generation of vapours. Contain and collect spillage with non-combustible absorbent material such as sodium bicarbonate, sodium carbonate, calcium carbonate, clean sand or non-acidic clay and then dampen the mixture with water. DO NOT USE peat

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moss or vermiculite. Sweep or scoop up using non-sparking tools and place into suitable containers for prompt disposal according to proper federal, state, and local regulations. Remove residual material with water.

7. HANDLING AND STORAGE

Handling: Contact with incompatible materials or exposure to temperatures exceeding the Self-Accelerating Decomposition Temperature (SADT) of 100°F (38°C) may result in a self-accelerating decomposition reaction with release of flammable vapours which may auto-ignite. Prevent product contamination. Keep away from heat, sparks, flames, and other ignition sources. Do not ingest, inhale, or expose eyes and skin to this product.

Emergency showers and eye wash stations should be readily accessible. Adhere to work practice rules established by government regulations. Do not cut, puncture, or weld on or near the container. Avoid breathing vapors and/or aerosols. Avoid contact with skin and eyes. Use only in well-ventilated areas. Use personal protective equipment. When using, do not eat, drink or smoke.

Storage: Keep containers tightly closed in a cool and well-ventilated place, away from incompatible materials. Store locked up and in corrosive resistant container or a container with a corrosion resistant inner liner. DO NOT store at temperatures exceeding 100°F (38°C) as this is the Self-Accelerating Decomposition Temperature (SADT). Compatible storage/contact materials for organic / ketone peroxides include 316 stainless steel, glass, polyethylene (PE), polytetrafluoroethylene (PTFE), and polypropylene (PP).

Incompatible Materials: Strong acids, Strong bases, Strong oxidizing agents, Reducing agents, Accelerators, Friedel-Crafts reaction catalyst, Transition metal salts, Metal ions, Rust, Copper, Brass.

8. EXPOSURE CONTROLS & PERSONAL PROTECTION

Exposure Limits :

Exposure limits have not been established for this product.

2-Butanone Peroxide

CAS # 1338-23-4

OSHA	Permissible Exposure Limit (PEL)	None Established
ACGIH	Threshold Limiting Value (TLV)	0.2 ppm (1.5 mg/m ³)
NIOSH	Recommended Exposure Limit (REL)	0.2 ppm (1.5 mg/m ³)

Dimethyl phthalate

CAS # 1317-11-3

OSHA	Permissible Exposure Limit (PEL)	5 mg/m ³
ACGIH	Threshold Limiting Value (TLV)	5 mg/m ³
NIOSH	Recommended Exposure Limit (REL)	5 mg/m ³

2-Butanone

CAS # 78-93-3

OSHA	Permissible Exposure Limit (PEL)/(STEL)	200 ppm/300 ppm
ACGIH	Threshold Limiting Value (TLV)	200 ppm (590 mg/m ³)
NIOSH	Recommended Exposure Limit (REL)/(STEL)	200 ppm/300 ppm

Hydrogen Peroxide

CAS # 7722-84-1

OSHA	Permissible Exposure Limit (PEL)	1 ppm (1.4 mg/m ³)
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ACGIH	Threshold Limiting Value (TLV)	1 ppm (1.4 mg/m ³)
NIOSH	Recommended Exposure Limit (REL)	1 ppm (1.4 mg/m ³)

Engineering Controls: Provide sufficient mechanical (general and/or local exhaust) ventilation to maintain exposure below permissible exposure limits. Provide readily accessible eye wash stations and safety showers.

Occupational Exposure Controls: Ensure adequate ventilation, especially in confined areas. Consider all potential hazards of this material, applicable exposure limits, job activities, and other substances in the work place when designing engineering controls and selecting PPE. Ensure that eyewash stations and safety showers are proximal to the work location. It is ultimately the responsibility of the employer to follow regulatory guidelines established by local authorities.

Protective and Hygiene measures: Do not inhale vapors. Wash hands before breaks and immediately after handling product. When using, do not eat, drink, or smoke. In case of clothes contamination, remove and wash contaminated clothing before re-use. Discard of contaminated leather articles.

Eye Protection: Recommended to wear chemical splash goggles at all times when using this product. Have a suitable eye wash station or bottle nearby in case of splashing into the eyes.

Hand Protection: Recommended to wear suitable resistant gloves and discard of gloves that show tears, pinholes, or signs of wear. Suitable gloves will be based on product use and the period of use, and may include neoprene, butyl-rubber, nitrile rubber, etc.

Skin Protection: Recommended to wear long-sleeved clothing, pants and proper foot covering in order to prevent direct skin contact with the product. If skin irritation develops, contact your facility health and safety professional or your local safety equipment supplier to determine the proper personal protective equipment for your use.

Respiratory Protection: A NIOSH-approved air-purifying respirator with an appropriate cartridge and/or filter may be appropriate under certain circumstances where airborne concentrations are expected to exceed exposure limits (if applicable) or if overexposure has otherwise been determined. Protection provided by air-purifying respirators is limited. Use a positive pressure, air-supplied respirator if there is any potential for uncontrolled release, exposure levels are not known or any other circumstances where an air-purifying respirator may not provide adequate protection.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Various Colored Liquid

Odor: Sweet

Odor Threshold: No data available

pH: No data available

Melting/freezing point: No data available

Boiling point: No data available

Boiling range: No data available

Flash point (Tag closed cup): 203°F (95°C)

Evaporation rate: No data available

Flammability: Lower Limit: No data available **Upper Limit:** No data available

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Vapor pressure: 5.2 mmHg @ 66°F (19°C)

Vapor density: No data available

Relative density: 1.01 g/cm³ (8.42 lb/gal) @ 68°F (20°C)

Solubility in water: Slightly soluble

Partition coefficient (n-octanol/water): No data available

Auto-ignition temperature: 167°F (75°C) (Self-Accelerating Decomposition Temperature (SADT); 45 gal.

Decomposition temperature: No data available

Viscosity (dynamic): 17.3 mPa*s @ 68°F (20°C)

10. STABILITY AND REACTIVITY

Reactivity: No decomposition if stored and applied as directed.

Chemical Stability: This material is chemically unstable and should only be handled under specific conditions.

Possibility of Hazardous Reactions: Hazardous polymerization does not occur, however this material does decompose and may release flammable vapours when in contact with heat above the published Self-Accelerating Decomposition Temperature (SADT).

Conditions to Avoid: Temperatures above the published SADT of 167°F (75°C)

Incompatible Materials: Strong acids, Strong bases, Strong oxidizing agents, Accelerators, Reducing agents, Friedel-Crafts reaction catalyst, Transition metal salts, Metal ions, Copper, Brass

Hazardous decomposition products: Hazardous organic compounds, Carbon monoxide, Carbon dioxide

11. TOXICOLOGICAL INFORMATION

Primary Routes of Exposure: Inhalation, Eye contact, Ingestion, Skin contact

Symptoms Related to Physical, Chemical and Toxicological Characteristics: Skin irritation, redness, burning sensation, severe skin damage, eye irritation, tearing of the eyes, severe eye damage, blindness, stomach or intestinal irritation, respiratory tract irritation, nose, throat and lung irritation, headache, nausea, dizziness, and vomiting.

Delayed and Immediate Effects & Chronic Effects from Exposure: The substance may have effects on the respiratory system. This substance is not a suspected carcinogen (see below).

Measures of Toxicity:

Acute toxicities are calculated based on component toxicities

Mixture: Acute Oral Toxicity: LD₅₀ Rat: 2,200 mg/kg

Acute Dermal Toxicity: LD₅₀ Rat: 1,700 mg/kg

Acute Inhalation Toxicity: LC₅₀ Rat: 9.0 mg/l

Carcinogen Claims:

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OSHA: No

International Agency for Research on Cancer (IARC): No; 4 [Probably not carcinogenic]

ACGIH: No; A5 [Not Suspected]

National Toxicology Program (NTP) Report on Carcinogens: No

12. ECOLOGICAL INFORMATION

Ecotoxicity: This substance is toxic to aquatic organisms. It is strongly advised that this substance does not enter the environment.

Persistence & Degradability: Based on component biodegradability, this product can be assumed to be readily biodegradable.

Bio-accumulative Potential: No data available

Mobility in Soil: No data available

Other Adverse Effects: No data available

13. DISPOSAL CONSIDERATIONS

Dispose of in accordance with Federal, State or Local regulations.

Contaminated packaging should be emptied as far as possible before disposal.

14. TRANSPORT INFORMATION

DOT SHIPPING CLASSIFICATION:

UN NUMBER: UN3105

PROPER SHIPPING NAME: Organic peroxide type D, liquid (Methyl ethyl ketone peroxide)

TRANSPORTATION HAZARD CLASS: 5.2

PACKING GROUP: II

HAZARD LABEL: 5.2

IMDG (Marine) SHIPPING CLASSIFICATION:

IMDG CODE: 5.2

UN NUMBER: UN3105

MARINE POLLUTANT: No

EmS: F-J, S-R

IMDG PACKING GROUP: II

HAZARD LABEL: 5.2

Description of the goods

ORGANIC PEROXIDE TYPE D, LIQUID (METHYL ETHYL KETONE PEROXIDE)

IATA (Air) SHIPPING CLASSIFICATION:

ICAO/IATA-DGR: 5.2

UN NUMBER: UN3105

HAZARD LABEL: Organic peroxide & Keep away from heat

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IATA-packing instructions – Passenger: 570

IATA -max. quantity – Passenger: 5L

IATA – packing instructions – Cargo: 570

IATA –max. quantity – Cargo: 10L

IAO packing group: II

Limited quantity Passenger: Forbidden

Description of the goods

Organic peroxide type D, liquid (Methyl ethyl ketone peroxide)

15. REGULATORY INFORMATION

All components of this product conform to the following national inventory requirements: US TSCA, EU EINECS and Canada DSL

SARA Title III

Section 302 – Extremely Hazardous Chemicals

The following ingredients are subject to the supplier notification requirements of Section 302 of the Superfund Amendments and Reauthorization Act (SARA/EPCRA) and the requirements of 40 CFR Part 37

Component	CAS #	Weight %
Hydrogen Peroxide	7722-84-1	< 1

Section 313 – Toxic Chemicals

The following ingredients are subject to the supplier notification requirements of Section 313 of the Superfund Amendments and Reauthorization Act (SARA/EPCRA) and the requirements of 40 CFR Part 37

None Listed

OTHER FEDERAL REGULATIONS

Components of this product are subject to RCRA Hazardous Waste requirements.

Clean Air Act (CAA) Hazardous Air Pollutants requirements and OSHA Process Safety Management (PSM) high hazard requirements.

CANADIAN REGULATIONS

Same as OSHA GHS Classification

STATE REGULATIONS

California Proposition 65

This product does not contain chemicals known to the state of California to cause cancer, birth defects, and other reproductive harm.

The components of this product may be included on the various state hazardous materials lists noted below.

California Hazardous Substances List/Permissible Exposure List

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California Toxic air contaminants
Connecticut Permissible Exposure Limits
Delaware List of Chemicals and RQs
Hawaii Permissible Exposure Limits
Idaho Toxic Air Pollutants
Illinois Toxic Air Contaminants List
Louisiana Toxic Air Pollutants
Maine Hazardous Air Pollutants
Maryland Toxic Air Pollutants for Existing Sources
Massachusetts Hazardous Substances List
Michigan Permissible Exposure Limits
Minnesota Hazardous Substances
Minnesota Permissible Exposure Limits
Nebraska Hazardous Air Pollutants
New Jersey RTK Hazardous Substances List/TCPA Extremely Hazardous Substances List
New York List of Hazardous Substances
Ohio Toxic Air Contaminants
Oklahoma Toxic Air Contaminants
North Carolina TAP Emissions Rates Requiring a Permit
Pennsylvania Hazardous Substances List
Rhode Island Toxic Air Contaminants
Tennessee Permissible Exposure Limits
Vermont Hazardous Air Contaminants/Permissible Exposure Limits
Washington Permissible Exposure Limits for Airborne Contaminants.
West Virginia Toxic Air Pollutant List
Wisconsin hazardous Air Contaminants

Note: Entries under Section 15 are not intended to be all inclusive of Federal and State laws and regulations. Please consult the appropriate agencies for further clarification of any requirements.

16. OTHER INFORMATION

Disclaimer: The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.